

CITE DEPARTMENT LOST AND FOUND MANAGEMENT SYSTEM

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ABSTRACT

This project proposes a desktop-based Lost and Found Management System for the CITE Department, Carmen Campus. The proposed system is intended to organize the reporting of lost items, record items that have been found, let users browse and search reported items, and keep a record of claim requests so that items can be returned to their rightful owners more reliably. The desktop application uses Java Swing for the graphical user interface and core functions, and MySQL, accessed through JDBC, for data storage. phpMyAdmin is used to create and administer the database during development and maintenance.

The project follows an Agile SDLC approach supported by Scrum practices. Development is divided into short sprints so that the team can present working features, collect feedback from the CITE Department, and improve the system before final deployment. This document also covers requirements, feasibility, risks, schedule, estimated cost, and Scrum activities.

I. PROJECT BACKGROUND

1.1 Client / Organization Background

The client of this project is the CITE Department of the Carmen Campus. As an academic department, it regularly encounters personal belongings that students, faculty, and visitors leave behind in classrooms, laboratories, corridors, and other campus areas, as well as inquiries from owners looking for items they have lost. The main users of the proposed system are the authorized CITE Department personnel who encode and monitor reports and claims. Students and faculty are the indirect beneficiaries, since they are the ones who report and recover items. The system is planned as a desktop application installed on the department's computer.

1.2 Current Process

Based on the process described by the project team, a person who loses an item may inform department personnel verbally or through a message. When an item is found and handed over, its details may be recorded informally or remembered by the person receiving it. When an owner asks about an item, personnel may need to check available notes or stored items. A centralized and consistently maintained record can make lost and found information easier to review.

1.3 Identified Problems

Based on the assumed process, the following problems may occur:

- Lost and found reports may be incomplete, forgotten, or never recorded.
- Owners may not know whether their item has already been turned in.
- Matching lost reports with found items depends on the memory of personnel and can be slow.
- It is difficult to verify who claimed an item and when it was claimed when no claim record exists.
- Summarizing how many items were reported, found, and claimed takes extra time when records are scattered or manual.

2. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of the project is to create a simple desktop-based system that helps the CITE Department record lost and found items, make those records easy to search and review, and keep track of claim requests so that items are returned in a more organized and accountable way.

2.2 Main Project Objective

To design and develop a desktop-based Lost and Found Management System for the CITE Department, Carmen Campus, that supports organized recording, searching, and claiming of lost and found items.

2.3 Specific Objectives

- To provide secure account registration and login for authorized users.
- To allow users to submit lost item reports with the item name, category, location, date, and description.
- To allow users to submit found item reports with the same set of details.
- To let users view and search lost items and found items and open the full details of a selected item.
- To allow a claim request to be submitted for a found item, including the claimant's name, email, and an additional message.
- To provide a records view of all claims and a dashboard showing totals of lost, found, and claimed items together with the most recent activity.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

User / Area	Included Functions
Registered User (CITE personnel)	Sign up, log in, view the dashboard, report a lost item, report a found item, view and search lost items, view and search found items, view item details, submit a claim request, view claim records, and log out.
Administrator	Log in, view the administrator dashboard, manage registered users, monitor lost and found item records, review claim requests, monitor claim statuses, view system reports, manage administrator profile information, and log out.
Dashboard	Display the total number of lost items, found items, and claims, and a list of the ten most recent lost, found, and claim entries with color-coded status.
System	Store users, lost items, found items, and claims in a MySQL database and validate required fields before saving.

3.2 Project Limitations

- The system is a desktop application and is not a web or mobile application, so it can only be used on computers where it is installed and connected to the database.
- The system depends on the available computer and local MySQL database configuration.
- Administrator and regular user functions are implemented, but access is limited to the functions provided by the current system.
- The system supports claim monitoring and different claim statuses; however, some advanced claim-processing features may still be improved in future versions.
- The system supports image information for lost and found item records, but image storage depends on the selected image file and local computer setup.
- The system supports email notifications for relevant system activities. SMS notifications are not included in the current version.

- Additional advanced report editing and deletion functions may be considered for future improvements.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The project is technically feasible for a student development team. The application is built with Java Swing, using the NetBeans GUI builder for the screens, and communicates with a MySQL database through the MySQL JDBC driver. phpMyAdmin is used to create, view, and maintain the database tables during development. All these tools run on a standard development computer and are sufficient for the reporting, search, and claim functions required by the project.

4.2 Operational Feasibility

The system is operationally feasible because it follows a simple flow that department personnel already understand: someone loses an item, someone finds an item, and an owner claims it. The interface uses a navigation panel (Home, View Lost Items, View Found Items, Claims Records, Logout), labeled forms, a category drop-down, and color-coded status values so that users can learn it with minimal training.

4.3 Economic Feasibility

The project is economically feasible because it uses free and open-source tools (Java, MySQL, phpMyAdmin) and the team provides the development labor. Possible expenses include transportation for client meetings, printing, backup storage, and minor hardware needs. These costs are small compared with the expected benefit of faster item recovery and better record keeping.

4.4 Project Charter Summary

Charter Item	Entry

Project Title	CITE Department Lost and Found Management System
Project Purpose	Improve how the CITE Department records lost items, found items, and claims through a desktop application.
Primary Users	CITE Department personnel (with students and faculty as indirect beneficiaries).
Major Deliverable	Working desktop system with documentation and a basic user guide.
Main Constraint	Limited development time, available hardware, and student-level resources.
Project Success	Users can report lost and found items, search and view them, and submit and review claims correctly using the system.

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
CITE Department Head / Administrator	Approves the project direction and provides the business requirements.
CITE Department Personnel	Main users who encode reports and claims and monitor records.
Students and Faculty	Report lost items, look for their belongings, and claim items; they are affected by the speed and accuracy of the process.
Development Team	Analyzes, designs, develops, tests, and documents the system.

5.2 Requirements Gathering Methods

- Client interview with CITE Department personnel to understand how lost and found items are currently handled.
- Observation of how items are received, stored, and returned in the department.
- Review of any existing notes, logbooks, or message threads used to record items.
- Follow-up consultation to confirm unclear requirements and the proposed screens.

5.3 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow a user to create an account using full name, email, and password.
FR-02	The system shall allow a registered user to log in using email and password.
FR-03	The system shall display a dashboard with the total lost items, found items, and claims, and the ten most recent reports.
FR-04	The system shall allow a user to submit a lost item report with item name, category, location lost, date lost, and description.
FR-05	The system shall allow a user to submit a found item report with item name, category, location found, date found, and description.
FR-06	The system shall validate that required fields are filled in and that dates follow the MM/DD/YYYY format before saving.
FR-07	The system shall list lost items and found items and allow searching by item name.
FR-08	The system shall display the full details of a selected lost or found item.
FR-09	The system shall allow a claim request for a found item with the claimant's name, email, and an optional message, saved with a Pending status.
FR-10	The system shall list all claim records and allow searching by item name.
FR-11	The system shall provide separate functions for regular users and administrators.
FR-12	The system shall allow administrators to view and manage registered user accounts.
FR-13	The system shall allow administrators to monitor lost and found item records.
FR-14	The system shall allow administrators to review claim requests and monitor their statuses.
FR-15	The system shall allow administrators to generate and view reports for lost items, found items, and claims.
FR-16	The system shall use password hashing to protect stored user passwords.
FR-17	The system shall support image information for lost and found item records.

5.4 Non-Functional Requirements

Category	Requirement
Usability	The interface should be simple, with consistent navigation and clear messages for missing or invalid input.
Performance	Common actions such as opening a list or searching by item name should respond within a reasonable time on the target computer.
Security	Access to the system requires login; passwords and database credentials should be protected before final deployment.
Reliability	Access to the system requires login; passwords and database credentials should be protected before final deployment.
Maintainability	Item categories and status values should be easy to update, and database access should be centralized so that connection settings can be changed in one place.

5.5 Sample User Stories and Acceptance Criteria

User Story	Sample Requirement
As a staff member	Report lost Item
As a staff member	Record Found Item
As a user	Search Items
As Claimant	Submit Claim
As a staff member, I want to report a lost item so that it is on record if someone turns it in.	Required fields are validated; a successful save shows a confirmation; the report appears in the lost items list with a Lost status.
As a staff member, I want to record a found item so that its owner can be located.	The item is saved with its details and appears in the found items list with a Found status.

As an administrator, I want to search items by name so that I can quickly check whether an item was reported.	Typing a keyword and searching shows only matching items.
As a claimant, I want to submit a claim for a found item so that my request is recorded.	Name and email are required; the claim is saved with today's date and a Pending status.
As an administrator, I want to see totals and recent activity so that I can monitor the lost and found process.	The dashboard shows counts for lost, found, and claims and the latest ten entries.

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

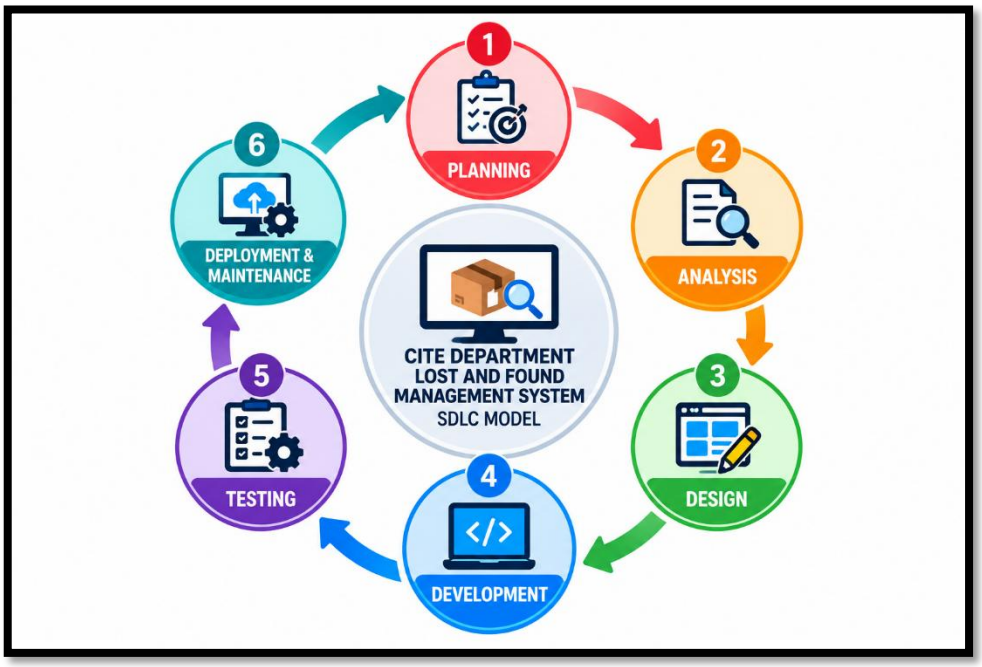
6.1 Development Methodology Used

The project uses an Agile development methodology supported by the Scrum framework. The work is divided into short sprints. In each sprint, the team selects priority tasks from the product backlog, develops and tests them, reviews the completed work with the client or adviser, and uses the feedback to improve the next sprint. This helps the team manage changes, track progress, and stay focused on the most important needs of the CITE Department.

6.2 SDLC Model Used

The selected SDLC model is Agile SDLC. Instead of completing the entire system in one long cycle, the system is built gradually through repeated iterations. Each iteration may include planning, requirements analysis, design, development, testing, and review, and feedback from the client or adviser is considered before the next iteration begins.

6.3 SDLC Model Figure



Figurev1. Agile SDLC Model Used for the Project

6.4 Software and Development Tools Used

The following software and technologies are planned for the development of the CITE Department Lost and Found Management System.

Software / Technology	Use in the Project
Java (Swing GUI, NetBeans GUI builder)	Used to build the desktop interface (login, sign-up, dashboard, report forms, item lists, detail views, and claim screens) and to implement the application logic.
MySQL with JDBC (MySQL Connector/J)	Used as the database for user accounts, lost items, found items, and claims. The application connects through JDBC.
phpMyAdmin	Used by the team to create, view, and maintain the MySQL tables and records during development and maintenance. It is an administration tool and not the user interface of the system.

6.5 Reason for Selecting the Methodology and SDLC Model

Agile SDLC with Scrum is appropriate because the department's needs become clearer once users see and try the actual screens. Short sprints let the team deliver the most important features first, such as login and reporting, demonstrate progress, receive feedback, and correct problems before moving on. This also suits the limited academic schedule by keeping work organized and visible.

6.6 Expected Development Approach

- Prepare and prioritize the initial system requirements.
- Divide the work into small features or backlog items.
- Develop a working set of features during each sprint.
- Test completed features before adding more work.
- Present progress to the client or adviser for feedback.
- Improve the system based on accepted feedback.

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stage	Application to the Lost and Found System
Planning	Identify the problem of unorganized lost and found handling, set objectives, users, scope, resources, and constraints.
Requirements Analysis	Interview and observe CITE personnel, list functional and non-functional requirements, and confirm item categories and claim rules.
System Design	Prepare the screen layouts, navigation flow, and the database structure for users, lost items, found items, and claims.
Development	Build sign-up and login, dashboard, lost and found reporting, item lists with search, item details, claim submission, and claim records.

Testing	Check each function, validate required fields and date formats, test invalid inputs, and have department personnel try the full report-to-claim workflow.
Deployment	Install the approved version and database on the department computer and prepare a basic user guide.
Maintenance	Correct errors, back up the database, and plan approved improvements such as claim approval and user roles.

8. PROJECT RISK MANAGEMENT

8.1 Possible Risks Discovered

The team identified events that may affect the project's schedule, cost, scope, or quality. The risks below are connected to this project.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
The department changes important requirements during development.	Medium	High	Confirm requirements every sprint and record approved changes before continuing.
Development takes longer than planned.	Medium	High	Prioritize core features, monitor weekly progress, and move low-priority tasks to later sprints.
Project files or the database are lost.	Low	High	Use source control and keep regular local and cloud backups, including exports of the database from phpMyAdmin.
Sensitive data is exposed (for example, stored	Medium	High	Store passwords in hashed form, use parameterized queries in all database operations, and use a dedicated database

passwords or unprotected database access).			account with a password instead of the default.
The application fails on the department computer (database not running, missing driver, or missing image files).	Medium	Medium	Document the setup, package required libraries and images with the application, and test on a computer similar to the client's before deployment.
A team member becomes unavailable.	Medium	Medium	Document work, share project access, and assign backup responsibilities.
Users find the workflow confusing.	Medium	Medium	Perform early user testing and simplify screens based on feedback.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The schedule shows the order of major tasks, their duration, and important milestones. It is a proposed schedule and should be adjusted to the actual academic calendar.

9.2 Sample Project Schedule

Activity	Duration	Target Period	Main Output
Planning and Client Study	1 week	Week 1	Problem, objectives, scope

Requirements Gathering	1 week	Week 2	Confirmed requirements
System and UI Design	2 weeks	Weeks 3-4	Wireframes and database design
Sprint 1 Development	2 weeks	Weeks 5-6	Sign-up, login, dashboard
Sprint 2 Development	2 weeks	Weeks 7-8	Lost and found reporting
Sprint 3 Development	2 weeks	Weeks 9-10	Item lists, search, details, claims, claim records
Testing and Revision	1 week	Week 11	Corrected release candidate
Deployment and Final Documentation	1 week	Week 12	Installed system and final documents

9.3 Project Milestones

Milestone	Target
Requirements approved	End of Week 2
Initial design approved	End of Week 4
Login and dashboard working	End of Week 6
Reporting and viewing working	End of Week 8
Feature-complete version	End of Week 10
Reporting and viewing working	End of Week 11
Feature-complete version	Week 12

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

Cost Item	Basis	Estimated Cost
Development Software	Free / open-source tools (Java, NetBeans)	PHP 0
Database Software	Free / open-source (MySQL, phpMyAdmin)	PHP 0
Transportation	Client meetings and testing visits at the Carmen campus	PHP 1,500
Printing / Documentation	Drafts, final copies, forms	PHP 600
Backup Storage / USB	Project backup	PHP 500
Hardware Adjustment	Allowance for minor peripherals or replacement needs	PHP 1,500
Contingency	Allowance for unexpected expenses	PHP 900
TOTAL ESTIMATED COST		**PHP 5,000**

10.2 Cost Justification

The budget focuses on development and deployment expenses rather than charging for student labor. Transportation and documentation support client meetings and project submission. Backup storage protects project files, while the hardware and contingency amounts provide an allowance for possible deployment needs.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	Responsibility
Product Owner	Represents the needs of the CITE Department, helps prioritize features, and confirms whether completed work meets the agreed requirements.

Scrum Master	Organizes Scrum activities, helps remove blockers, and keeps the team focused on the sprint goal.
Development Team	Designs, codes, tests, documents, and improves the system.

11.2 Product Backlog

Priority	Backlog Item	Expected Result
High	Sign-up and login	Authorized users can create an account and access the system.
High	Report lost item	A lost item report is validated and saved.
High	Report found item	A found item report is validated and saved.
High	View lost and search items	Users can browse items and open their details.
High	Submit claim	A claim request can be submitted and recorded.
High	User roles	Regular users and administrators have separate system functions.
High	Claim Management	Administrators can review claims and monitor claim statuses.
Medium	User Management	Administrator can review and manage registered users.
Medium	Dashboard	The system displays item and claim counts and recent activities.
Medium	Administrative reports	Administrators can generate lost, found and claim reports.
Medium	Password Security	User passwords are protected using password hashing.

11.3 Sprint Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Establish access and navigation.	Sign-up, login, home dashboard, navigation panel.

Sprint 2	Complete the reporting process.	Sign-up, login, home dashboard, navigation panel.
Sprint 3	Complete viewing and claiming.	Item lists, search, item details, claim form, claim records.
Sprint 4	Prepare the system for final use.	Testing, bug fixes, user feedback, security improvements, deployment preparation.

11.4 Scrum Events

Scrum Event	How the Team Applies It
Sprint Planning	Select backlog items that can realistically be completed and agree on the sprint goal.
Daily Scrum / Team Check-in	Briefly discuss completed work, current tasks, and blockers.
Sprint Review	Demonstrate completed functions to the client, adviser, or project group and gather feedback.
Sprint Retrospective	Discuss what worked, what caused problems, and what to improve in the next sprint.

11.5 Simple Scrum Progress Tracking

The team uses a simple task board with three columns: To Do, In Progress, and Done. Each backlog item has an assigned member and status. At the end of every sprint, the team compares planned work with completed work and uses the result to improve the next sprint plan.

12. PROJECT SUMMARY

The CITE Department Lost and Found Management System provides an organized desktop-based solution for recording, searching, viewing, and claiming lost and found items.

The system supports regular users and administrators with separate functions based on their responsibilities. Regular users can report lost and found items, search records, view item details, and submit claim requests. Administrators can manage registered users, monitor lost and found records, review claim requests, monitor claim statuses, and generate system reports.

The system uses Java Swing for the user interface and MySQL through JDBC for database storage. It also applies password hashing to improve account security and supports image information for lost and found item records. Future improvements may include automatic email notifications and additional advanced editing and deletion functions. These improvements can further enhance communication, flexibility, and overall system usability.